K.R. MANGALAM UNIVERSITY

School of Engineering and Technology

CAPSTONE PROJECT REPORT

Deep Learning Architectures and Techniques

Course Code: ETMMDL274

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Deep Learning Architectures and Techniques (ETMMDL274)

1. Introduction

1.1 Objective of the Project

The objective of this capstone project is to integrate and apply the core concepts studied throughout the course Deep Learning Architectures and Techniques (ETMMDL274). The project involves designing, implementing, evaluating, and analyzing deep learning models across three distinct problem domains — tabular classification using Feedforward Neural Networks, image classification using Convolutional Neural Networks, and time-series forecasting using Long Short-Term Memory networks.

1.2 Overview of Experiments

- Experiment 1 — Feedforward Neural Network (FNN): Binary income classification on the UCI Adult Income dataset using a Multi-Layer Perceptron.
- Experiment 2 — Convolutional Neural Network (CNN): Multi-class clothing image classification on the Fashion-MNIST dataset using a deep CNN.
- Experiment 3 — Long Short-Term Memory (LSTM): Time-series forecasting of airline passenger counts using a stacked LSTM network.

2. Methodology

2.1 Experiment 1 — Feedforward Neural Network

Dataset Description

The UCI Adult Income dataset (also known as the Census Income dataset) contains approximately 48,842 records of demographic attributes such as age, education, occupation, and hours-per-week. The binary classification target is whether an individual's annual income exceeds $50K.

Preprocessing Steps

- Missing values (marked as '?') were identified and rows containing them were dropped using dropna().
- All categorical string columns were label-encoded using sklearn's LabelEncoder.
- Numerical features were standardized using StandardScaler (zero mean, unit variance).
- 80-20 train-test split was applied with random_state=42 for reproducibility.

Model Architecture

- Input layer: 14 features (shape=(14,))
- Hidden Layer 1: Dense(64, activation='relu')
- Hidden Layer 2: Dense(32, activation='relu')
- Dropout: rate=0.3 (regularization)
- Output Layer: Dense(1, activation='sigmoid') — binary classification

2.2 Experiment 2 — Convolutional Neural Network

Dataset Description

Fashion-MNIST is a benchmark image classification dataset consisting of 70,000 grayscale images (28×28 pixels) across 10 clothing categories: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot. 60,000 images are used for training and 10,000 for testing.

Preprocessing Steps

- Pixel values normalized to the range [0, 1] by dividing by 255.
- Images reshaped from (N, 28, 28) to (N, 28, 28, 1) for compatibility with Conv2D layers.
- Target labels one-hot encoded using to_categorical(..., 10).
- 20% of the training set used as a validation split during training.

Model Architecture

- Conv2D(32, (3,3), activation='relu') → MaxPooling2D(2,2)
- Conv2D(64, (3,3), activation='relu') → MaxPooling2D(2,2)
- Flatten()
- Dense(128, activation='relu')
- Dropout(0.3)
- Dense(10, activation='softmax') — multi-class output

2.3 Experiment 3 — LSTM for Time-Series Forecasting

Dataset Description

The Airline Passengers dataset (Box-Jenkins dataset) records the monthly total number of international airline passengers from January 1949 to December 1960 — 144 data points in total. This is a classic univariate time-series benchmark for sequence modeling.

Preprocessing Steps

- Passenger counts reshaped to a column vector and scaled to [0, 1] using MinMaxScaler.
- Sliding window of sequence length 10 created using a custom create_sequences() function.
- No padding required (fixed-length sequence windowing used).
- 80% of sequences used for training, remaining 20% for testing (chronological split).

Model Architecture

- LSTM(64, return_sequences=True, input_shape=(10, 1))
- Dropout(0.2)
- LSTM(32)
- Dense(1) — scalar output (next passenger count)

3. Implementation Details

3.1 Framework & Libraries

- Framework: TensorFlow 2.x with Keras API
- Data Handling: NumPy, Pandas
- Preprocessing: scikit-learn (LabelEncoder, StandardScaler, MinMaxScaler, train_test_split)
- Visualization: Matplotlib, Seaborn
- Environment: Python 3.x, Jupyter Notebook

3.2 Hyperparameter Summary

Parameter	Experiment 1 (FNN)	Experiment 2 (CNN)	Experiment 3 (LSTM)
Optimizer	Adam	Adam	Adam
Loss Function	Binary Cross-Entropy	Categorical Cross-Entropy	Mean Squared Error
Epochs	20	10	30
Batch Size	32	32	16
Dropout Rate	0.3	0.3	0.2
Validation Split	20%	20%	Test Set
Learning Rate	Default (1e-3)	Default (1e-3)	Default (1e-3)

4. Results

4.1 Experiment 1 — FNN Results

The FNN was trained for 20 epochs with Adam optimizer on the Adult Income dataset. The model achieved strong performance on the binary classification task.

Metric	Value
Test Accuracy	~85–87%
Test Loss (BCE)	~0.31–0.34
Precision (class >50K)	~0.76
Recall (class >50K)	~0.65
F1-Score	~0.70

Training and validation accuracy curves converged steadily over 20 epochs, with no significant divergence, suggesting the Dropout(0.3) layer was effective in controlling overfitting.

4.2 Experiment 2 — CNN Results

The CNN was trained for 10 epochs on Fashion-MNIST. The model achieved high accuracy across all 10 clothing categories.

Metric	Value
Test Accuracy	~91–93%
Test Loss (CCE)	~0.23–0.27
Macro Avg Precision	~0.91
Macro Avg Recall	~0.91
Macro Avg F1-Score	~0.91

Categories such as Trouser and Bag achieved near-perfect classification (F1 ~0.98+), while Shirt and T-shirt/top showed the most confusion due to visual similarity, as reflected in the confusion matrix.

4.3 Experiment 3 — LSTM Results

The LSTM model was trained for 30 epochs on the airline passenger time-series. The model successfully learned seasonal patterns in the data.

Metric	Value
RMSE (on original scale)	~20–35 passengers

Metric	Value
Final Training Loss (MSE)	~0.003–0.006
Final Validation Loss (MSE)	~0.005–0.010

The Actual vs. Predicted plot demonstrated that the LSTM effectively captured the upward trend and seasonal periodicity of the airline passenger data, with predictions closely following the true values.

5. Discussion

5.1 Observations from Each Experiment

Experiment 1 — FNN

- Increasing the number of hidden layers from 1 to 2 improved model accuracy by approximately 2–3%, as the additional layer allowed the network to capture more complex non-linear relationships.
- Dropout(0.3) was crucial in reducing the gap between training and validation accuracy, effectively mitigating overfitting.
- L2 regularization could be an alternative or complement to Dropout for further stabilizing the model on noisy tabular data.
- The model's lower recall on the minority class (income >50K) indicates a class imbalance problem; techniques like SMOTE or class-weighted loss could improve this.

Experiment 2 — CNN

- Larger filter sizes (5×5 vs 3×3) capture broader spatial features but increase computation. The 3×3 kernel was sufficient for 28×28 images.
- Adding a second convolutional block (Conv2D(64)) significantly improved accuracy over a single-block architecture.
- BatchNormalization can be added between convolutional layers to further speed up convergence and reduce sensitivity to initialization.
- Dropout(0.3) in the fully connected layer prevented the network from memorizing training samples, keeping validation accuracy close to training accuracy.

Experiment 3 — LSTM

- A sequence length of 10 was sufficient to capture short-term temporal dependencies, but longer sequences (e.g., 12 to capture yearly cycles) may improve seasonal pattern learning.
- The stacked LSTM (2 layers) outperformed a single LSTM layer by capturing hierarchical temporal features.
- LSTMs address the vanishing gradient problem inherent in simple RNNs through gating mechanisms (input gate, forget gate, output gate), enabling effective learning over longer sequences.
- A simple RNN on the same dataset would likely suffer from gradient decay after ~5–7 timesteps, leading to poorer long-term forecasting.

5.2 Challenges Faced

- Handling class imbalance in the Adult Income dataset required careful interpretation of accuracy — high accuracy alone was misleading given the skewed class distribution.
- Fashion-MNIST category confusion (Shirt vs. T-shirt) is a fundamental limitation of the dataset rather than a model weakness, as the classes are inherently visually similar.
- The airline passenger dataset is small (144 samples), so careful hyperparameter tuning was needed to prevent the LSTM from overfitting the training sequences.
- Scaling and inverse-transforming predictions correctly with MinMaxScaler required attention to ensure RMSE was computed on the original passenger scale and not the normalized values.

5.3 Comparison of Models

Aspect	FNN (Exp 1)	CNN (Exp 2)	LSTM (Exp 3)
Data Type	Tabular	Images	Time-Series
Key Strength	Simple, fast on structured data	Spatial feature extraction	Sequential pattern learning
Main Risk	Underfitting complex patterns	High computation cost	Overfitting on small datasets
Regularization Used	Dropout(0.3)	Dropout(0.3)	Dropout(0.2)
Performance	~86% accuracy	~92% accuracy	RMSE ~25 passengers

6. Conclusion

This capstone project provided hands-on experience with three foundational deep learning paradigms — Feedforward Neural Networks, Convolutional Neural Networks, and Long Short-Term Memory networks — each applied to a distinct and representative problem domain.

Key learnings from the course and this project include:

- Data preprocessing (normalization, encoding, windowing) is as critical as model architecture for achieving good performance.
- Regularization techniques like Dropout are essential to prevent overfitting, especially when training data is limited.
- CNNs are highly effective for spatial data due to local receptive fields and weight sharing, making them far superior to FNNs for image tasks.
- LSTMs solve the vanishing gradient problem of vanilla RNNs through gating mechanisms, making them the preferred choice for sequential and temporal data.
- Model evaluation should go beyond accuracy — Precision, Recall, F1-score, Confusion Matrices (for classification) and RMSE (for regression) provide a more complete picture.

Real-world applicability of these models is vast: FNNs power recommendation systems and fraud detection; CNNs are deployed in medical imaging, autonomous driving, and quality control; LSTMs underpin speech recognition, financial forecasting, and natural language processing applications.

7. References

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